Term 4 Checklist

Topic	Learning Outcomes	RAG Yourself	
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Energy	Name the different forms of energy store e.g. kinetic, thermal		
	Draw energy transfer diagrams for a candle, a catapult and a TV		
	Identify useful and wasted energy in an energy transfer		
	Calculate the percentage efficiency of an energy transfer		
	Describe the difference in terms of energy transfer of an efficient device and a less efficient device		
	Explain how fossil fuels are used to generate electricity in a power station		
	Define what a renewable energy resource is		
	State examples of renewable energy resources e.g. solar, wind		
	Explain how solar, wind and tidal resources work in terms of energy transfer		
	Describe how human activities have affected the carbon cycle		
	Describe how carbon dioxide causes a greenhouse effect		
	Explain how human actions have contributed towards global warming		
	List some impacts of global warming		
	Name the three most common gases in our atmosphere		
Chemicals	State the approximate percentage of each gas in our atmosphere		
	Describe what an element is in terms of types of atoms		
	Describe how a molecule is formed from atoms		
	Compare chemical compounds to mixtures		
	Explain why chemical compounds are hard to separate		
	Draw particle diagrams of the three states of matter: solids, liquids and gases		
	State the physical properties of the three states of matter		
	Compare the kinetic energy of particles in different states of matter		
	Explain why solids and liquids cannot be compressed easily, but gases can		
	Identify the four main changes of state		
	Describe what happens to the particles in a substance during a change of state		
	List a range of common household acids and bases		
	Describe how litmus paper can be used to test acids and bases		
	Describe how universal indicator can be used to test acids and bases		
	Match colours to numbers on the pH scale		
	Compare how useful different pH indicators are		
	Describe what a neutralisation reaction is		
	Give the chemical equation for a neutralisation reaction		
	Identify common scientific warning labels e.g. irritant, corrosive, toxic		
	Define the key terms; solution, solvent, solute and soluble		
	Draw a particle diagram to show how a solution is formed		

	Describe how to separate a soluble solid from a solution by evaporation		
	Explain why filtration could not be used to separate a mixture of salt and water		
	Describe how distillation can be used to separate a mixture of liquid		
	State the method of separation which is used for a mixture of coloured dyes		
	Describe a method to identify the coloured dyes which have gone into a paint mixture		
	Explain why chromatography cannot be used to separate the pigments in a colouring pencil		
	Identify which separation technique would be suitable for a given mixture		
Forces and Motion	Describe the difference between contact and non-contact forces		
	State a range of contact and non-contact forces		
	State which variable goes on each axis		
	Draw force arrows on objects that are at rest or moving		
	Calculate the resultant force of an object when two forces are acting on it		
	Explain why an object will start to accelerate using the idea of forces		
	Explain why an object will travel at a steady speed using the idea of forces		
	State the units of mass and weight		
	Explain why your weight changes on different planets but your mass does not		
	"Calculate your mass or weight. Try this one: Earth's gravity is 9.8N/kg, Mr Wormstone has a mass of 78kg. What is his weight?"		
	State the forces that occur when a boat moves through water and floats on water		
	State useful and unhelpful examples of friction		
	Explain why friction occurs between two surfaces		
	Describe the difference between the independent and dependent and control variables		
	State the equation that links speed, distance and time.		
	State the units of speed, distance and time		
	"Calculate speed, distance or time using the equation s = d/t. Try this one: A car travels 90m in 45s. What is its speed?"		
	Describe factors that affect stopping distances		
	Describe the difference between braking distance and thinking distance		
	Explain the difference between speed (or velocity) and acceleration		
	Recall and use a formula to calculate acceleration		
	Explain how pressure is caused inside a can of coke in terms of particles		
	Use the idea of forces why a rocket can take off		
	Explain how more aerodynamic shapes help rockets fly		